Is Quantum Computing Ready for Real-Time Database Optimization?

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I. Background/Question/Methods.

Database systems encompass several performance-critical optimization tasks, such as join ordering (e.g., [1]–[8]) and index tuning (e.g., [9]–[11]). As data volumes grow and workloads become more complex, these problems have become exponentially harder to solve efficiently [12]. Quantum computing, especially quantum annealing [13], is a promising paradigm that can efficiently explore very large search spaces [14], [15] through quantum tunneling [16]. As illustrated in Fig. 1, it can escape local optima by tunneling through energy barriers rather than climbing over them. Earlier works [17]–[24] mainly focused on providing an abstract representation (e.g., Quadratic Unconstrained Binary Optimization (QUBO) [25]) for the database optimization problems (e.g., join order [1]) and overlooked the real integration within database systems due to the high overhead of quantum computing services (e.g., a minimum 5 s runtime for D-Wave's CQM-Solver [26]). Recently, quantum annealing providers offered more low-latency solutions, e.g., NL-Solver [27], which paves the road to actually realize quantum solutions within DBMSs [28]. However, this raises new systems research challenges in balancing efficiency and solution quality.

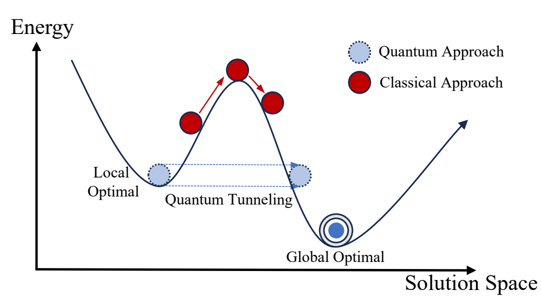

Fig. 1: Quantum Annealing Compare with Classical Approach.

In this talk, we show that this balance is possible to achieve. As a proof of concept, we present Q^2O, the *first* real Quantum-augmented Query Optimizer. Fig. 2 shows the end-to-end workflow: we encode the join order problem as a nonlinear model, a format solvable by the NL-Solver, using actual database statistics; the solution is translated into a plan hint that guides PostgreSQL's optimizer to produce a complete plan. Q^2O is capable of handling actual queries in real time.

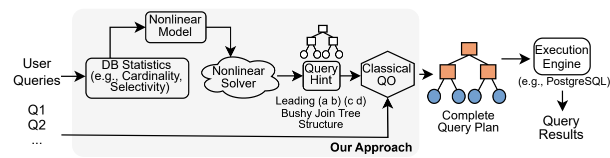

Fig. 2: The Workflow of our proposed Q^2O Framework.

II. Results/Conclusions

We set up PostgreSQL 16.4 with the corresponding *pg_hint_plan* version. We evaluate Q^2O using the *Join Order Benchmark (JOB)* [29] as real-word optimization challenges. **Query Plan Quality.** We evaluate the quality of the generated query plan by measuring the execution time for the JOB workload on PostgreSQL. Fig. 3 compares our plans with PostgreSQL's default query plans. Among 113 queries, our method yields speedups on 31 queries, with a maximum latency reduction of 92.7% and an average reduction of 42.09% on those queries.

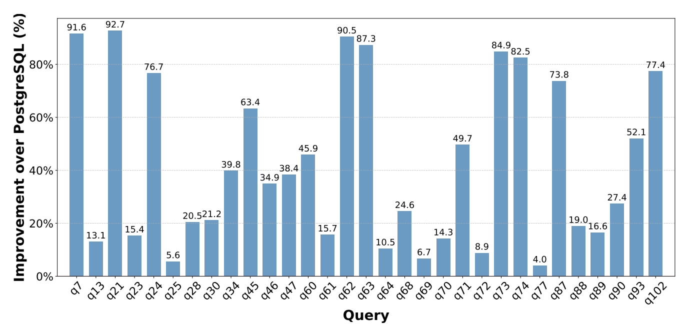

Fig. 3: A Selection of Query Plans Generated by Our Approach Outperforms PostgreSQL's Default Plans.

End-to-End (E2E) Latency. We measure E2E latency over the full pipeline. For PostgreSQL, it comprises planning time and execution time; for Q^2O, it includes PostgreSQL planning (with hints), PostgreSQL execution, and NL-Solver time. Table I reports per-component latencies for four JOB queries.

TABLE I: End-To-End Latency Breakdown and Comparison Between Our Approach and PostgreSQL.

	Time (ms)	q21	q60	q62	q63
PG	Planning	1.00	3.14	3.17	1.16
	Execution	3581.40	10951.31	4680.26	4846.42
Ours	Planning with hints	2.24	8.31	9.03	9.02
	Execution	272.35	6010.00	429.58	585.60
	NL-Solver	2530.22	2748.32	2854.31	2816.42
Gain	**Exec Time ↑**	**13.15x**	**1.82x**	**10.89x**	**8.28x**
	E2E Latency ↑	**1.28x**	**1.25x**	**1.42x**	**1.42x**

Although the NL-Solver introduces additional overhead (primarily unavoidable cloud communication), our method produces higher-quality join order solutions, yields execution-time speedups up to $13.15\times$, and reduces overall E2E latency.

In conclusion, we present Q^2O, which leverages quantum computing to solve database optimization in real-time settings. Our evaluation demonstrates speedups in both query execution time and end-to-end latency. Looking ahead, this framework shows potential to tackle larger-scale and complex database optimization problems in real-time scenarios.

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